

# Suraj Ranganath

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## Education

### University of California San Diego

San Diego, CA

M.S. in Data Science (GPA: 4.0/4.0)

Sept 2025 – June 2027

Courses: Safety in Generative AI, Web Mining and Recommender Systems, Advanced Text Mining, Statistical Models

### Vellore Institute of Technology

Vellore, India

B.Tech. in Computer Science and Engineering (GPA: 3.98/4.0, Top 5 in class)

Aug 2020 – May 2024

Courses: Machine Learning, Linear Algebra, Artificial Intelligence, Data Structures and Algorithms, Database Management

## Experience

### Causal Intelligence Lab, UC San Diego

Dec 2025 – Present

Research Assistant

San Diego, CA

- Working with Prof. Huang on extending memory for **video diffusion models** toward **embodied world modeling**.

### BalkanID

May 2023 – Jun 2025

Machine Learning Engineer

Remote

- Built BalkanID Copilot, an **agentic identity security system** leveraging **LLMs** and **knowledge graphs**, adopted by 6+ paying customers & earning over **\$200K in revenue** by surfacing actionable risk insights and cutting compliance reporting time by 80%.
- Designed and productionized a text-to-Cypher stack by LORA fine-tuning and layering RAG over a multi-tenant **Identity Intelligence Graph** (5M+ entities/relations), enabling natural-language querying of IAM policies and entitlements.
- Architected and deployed an **MCP server** with connector integrations, enabling natural language commands to trigger actions and workflows across the platform and 20+ enterprise services.
- Built multi-tenant **ETL pipelines** ingesting ~50GB/tenant from 100+ enterprise sources across 20+ identity & risk data types; normalized, deduped, and entity-resolved feeds into our identity graph to power analytics, compliance, and ML features.
- Conceived and engineered the BalkanID IAM Risk Analyzer, applying graph-based **role mining** and just-in-time access provisioning to enforce least privilege, reduce identity risk exposure, and accelerate access approval turnaround time by **2-3X**.
- Owned **model reliability** for Copilot: built evaluation harnesses and production monitoring for drift and behavior regressions, enabling 3-4x faster iteration on new LLM and retrieval variants through safe rollouts backed by offline evals and live telemetry.
- Co-inventor on **4 U.S. patents in agentic AI** and **graph machine learning** for identity security - delivering actionable risk insights, cutting manual reviews, and enabling autonomous governance workflows.

### Indian Institute of Technology, Bombay

Jan 2023 – Jun 2023

Research Intern

Mumbai, India

- Collaborated with Prof. Mitra, EdTech Lab on **multimodal data analysis** for spatial problem-solving research, in 6-person team.
- Trained **MLP classifier** to identify data-backed signatures of 5 spatial thinking strategies, informed by prior qualitative studies.
- Designed and managed multimodal data acquisition (EEG, eye tracking, GSR, affective video analysis), producing a novel dataset of **300+ hours**, published in **Nature Scientific Data**.

## Technical Skills

**Languages:** Python, Cypher, SQL, Java, Go, Javascript, Typescript, React, NoSQL, Matlab, R, C, C++, Bash, Shell, Linux  
**Machine Learning:** PyTorch, Agentic Frameworks, RAG, MCP, A2A, LLM Post-Training, LLM Monitoring and Evaluation  
**Software and Frameworks:** FastAPI, Django, Pandas, Matplotlib, NLTK, PySpark, PyTorch Geometric, Liger Kernel  
**Tools and Platforms:** Git, Docker, Kubernetes, AWS, Traefik, ArgoCD, GraphQL, Selenium, Tinker, HuggingFace, Axolotl  
**Databases:** Neo4J, MySQL, Qdrant, Pinecone, PostgreSQL, MongoDB, Redis, Firebase, SQLite3

## Patents and Publications

- Knowledge Graph-Enhanced AI Copilot Platform for Intelligent Identity Security Governance and Lifecycle Management. (2025).** Ranganath, S., Raghavendra, A., *et al.* U.S. Patent 19/055,635, Published Nov 6, 2025. Patent pending.
- A Multisensor Dataset of South Asian Post-Graduate Students Working on Mental Rotation Tasks. (2025).** TS, A., Ranganath, S., *et al.* Nature Scientific Data, 12(1), 563.
- Recommendation and Remediation of Role-Based Access Control Postures for Identities. (2025).** Ranganath, S., Grewal, D., *et al.* U.S. Patent 19/195,207, filed April 30, 2025. Patent pending.
- Multi-Agent Identity Governance and Administration System. (2025).** Raghavendra, A., Ranganath, S., *et al.* U.S. Patent 19/194,684, Published Nov 6, 2025. Patent pending.
- Method for Provisioning and De-Provisioning Just-in-Time, Purpose-Based Access for Identities within Applications. (2025).** Wong, N., Deep, A., *et al.* U.S. Patent 19/194,524, Published Nov 6, 2025. Patent pending.

## Projects

### StealthRL: Fairness-Aware GRPO Post-Training for AI-Text Detector Red-Teaming

Dec 2025

- Built an **RL post-training** pipeline to learn a paraphrasing policy that generalizes across multiple AI-text detector families while preserving semantic meaning and fluency, with a focus on **ESL fairness**.
- Implemented **GRPO** with LoRA adapters on Qwen3-4B-Instruct and a composite reward (detector-ensemble score, semantic similarity, perplexity, ESL fairness penalty); ran training and ablation studies on **Tinker**.

### Model Organisms for Privacy & Backdoor Failures in Image Classifiers

Sep 2025

- Implemented the **membership inference** framework on Fashion-MNIST: trained shadow models and an **XGBoost** attack classifier, plus a confidence-based baseline, under a standard threat model where the adversary only observes model outputs.
- Empirically showed that for a well-regularized CNN, both attacks collapse to near-random performance (**ROC AUC  $\approx$  0.5**), then trained **DP-SGD** variants to probe the privacy-utility tradeoff as test accuracy falls from  **$\sim$ 90% to  $\sim$ 65%** under stronger privacy.
- Implemented a **backdoor poisoning** on MNIST: injecting a subtle Pikachu trigger into just **1.75%** of training data yields **100%** attack success with **95.69%** clean accuracy, illustrating how catastrophic misbehavior can hide behind strong aggregate metrics.